

Day 16: Advanced DAX Iterators (X-Functions)

📌 Definition to Remember

Iterator Functions (X-Functions) like **SUMX**, **AVERAGEX**, and **RANKX** evaluate an expression row-by-row over a specified table, creating a Row Context before aggregating the results.

1. **SUMX()** vs **SUM()**

- SUM(Table[Column])** only takes a single physical column.
- SUMX(Table, Expression)** calculates a dynamic expression for every row and then sums:

```
Total Net Revenue =  
SUMX(  
    Fact_Sales,  
    Fact_Sales[Quantity] * Fact_Sales[UnitPrice] * (1 - Fact_Sales[Discount])  
)
```

2. Dynamic Ranking: **RANKX()**

```
Customer Sales Rank =  
RANKX(  
    ALL(Dim_Customer[CustomerName]),  
    [Total Sales],  
    , // Optional value  
    DESC,  
    Dense  
)
```

3. DAX Variables: **VAR ... RETURN**

Variables improve performance (calculated once and cached) and enhance readability:

```
Executive KPI Metric =  
VAR CurrentSales = [Total Sales]  
VAR PriorSales   = [Sales PY]  
VAR Variance     = CurrentSales - PriorSales  
VAR GrowthPct    = DIVIDE(Variance, PriorSales, 0)  
RETURN  
    IF(CurrentSales >= PriorSales, "▲ " & FORMAT(GrowthPct, "0.0%"), "▼ " & FORMAT(GrowthPct, "0.0%"))
```

4. **SELECTEDVALUE()**

Returns the single value currently selected in a slicer, or a default fallback if multiple items are selected:

```
Dynamic Chart Title = "Sales Analysis for " & SELECTEDVALUE(Dim_Region[RegionName], "All Regions")
```

★ Must-Master Skills & Interview Points

1. Why **VAR** prevents duplicate query execution in complex branching DAX expressions.
2. How **RANKX** calculates dynamic ranks across table subsets.
3. When to use **SELECTEDVALUE()** for dynamic card titles and parameter cards.

⚡ Quick Recall

SUMX/AVERAGEX (Row-by-row iterators) + RANKX (Dynamic ranking) + VAR/RETURN (Cached clean variables)